

Solution Requirements

Date: 30 September 2026

Team ID: SWTID-2026-3656

Project Name: EduGenie – Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	Serve the EduGenie homepage	Must	GET <code>/</code> returns rendered <code>index.html</code>
FR-02	Explain a concept locally	Must	<code>/explain</code> uses LaMini-Flan-T5 Seq2Seq generation
FR-03	Answer student questions	Must	<code>/ask</code> returns a Gemini-generated answer
FR-04	Generate a quiz	Must	<code>/quiz</code> validates five questions and four options
FR-05	Summarize text	Must	<code>/summarize</code> returns a student-friendly summary
FR-06	Generate a learning path	Must	<code>/learning-path</code> includes four required sections
FR-07	Handle empty input	Must	Useful validation error is returned
FR-08	Handle API/model errors	Must	Safe response without secret exposure
FR-09	Use asynchronous UI requests	Should	Fetch updates result cards without reload
FR-10	Support responsive UI	Should	Layout works on desktop and mobile

Non-Functional Requirements

- Python 3.10+ compatible.
- Modular beginner-readable code.
- Cached local model loading.
- Environment-based configuration.
- Responsive, accessible, and reduced-motion-aware UI.